Revision | **Covers** 4-1 → 4-5 | **File** 8 / 11

DT

Which rule do I use?

كيف أختار القانون الصحيح خلال عشر ثوانٍ

Run this in order on every question. It takes ten seconds and it is the difference between a confident answer and a guess.

Q1 **Does the question contain "not", "neither", "none", or "at least"?**

→ Yes: use the complement, `1 − P(opposite)`. Compute the easy side. This check comes first because it can collapse the whole problem.

Q2 **Does it say "given that", "if", "of those who", or state a fact as already settled?**

→ Yes: conditional probability, `P(B|A) = P(A∩B)/P(A)`. From a table, the condition's row or column total is the denominator.

Q3 **Does it say "or"?**

→ Addition rule. Then ask: can both happen together? No → `P(A)+P(B)`. Yes → `P(A)+P(B)−P(A∩B)`.

Q4 **Does it say "and", or describe a sequence of stages?**

→ Multiplication rule. Then ask: with replacement / separate devices? → independent, `P(A)·P(B)`. Without replacement / "of those"? → dependent, `P(A)·P(B|A)`.

Q5 **Is the sample space too big to list?**

→ Counting rules. Then ask: does order matter? Yes → ${}_nP_r$. No → ${}_nC_r$. Several categories combined → multiply them (Fundamental Counting Rule).

Q6 **Is it asking for a probability after all that counting?**

→ `P(E) = n(E)/n(S)`. Count the event and the sample space with the same kind of rule — never mix a permutation numerator with a combination denominator.

خطوات اختيار القانون بالعربي — طبّقها بالترتيب

1. هل في السؤال **at least** / **none** / **neither** / **not**؟ ← استخدم المتمّمة (العكس) `P(العكس) = 1 −` .
2. هل فيه **given that** / **if** / **of those who**، أو حقيقة وقعت مسبقًا؟ ← احتمال شرطي، والشرط في المقام.
3. هل فيه **or**؟ ← قاعدة الجمع. هل يمكن وقوعهما معًا؟ لا ← اجمع. نعم ← اجمع واطرح التقاطع.
4. هل فيه **and** أو مراحل متتالية؟ ← قاعدة الضرب. مع إرجاع ← اضرب مباشرة. بدون إرجاع ← الاحتمال الثاني يتغيّر.
5. هل فضاء العينة أكبر من أن يُسرد؟ ← قواعد العدّ. الترتيب مهم ← `nPr` . غير مهم ← `nCr` . عدة فئات ← اضرب.
6. هل المطلوب احتمال بعد العدّ؟ ← `n(E) ÷ n(S)` ، بنفس نوع العدّ في البسط والمقام.

THE ONE CONSISTENCY RULE THAT CATCHES MOST §4-5 ERRORS

Whatever you use for `n(S)`, use for `n(E)`. If the sample space counts *unordered* selections (combinations), the event must count unordered selections too. A numerator built from permutations over a denominator built from combinations is the most common wrong answer in the whole chapter, and it is always too large by a factor of `r!`.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use? ← this file

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief

version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.